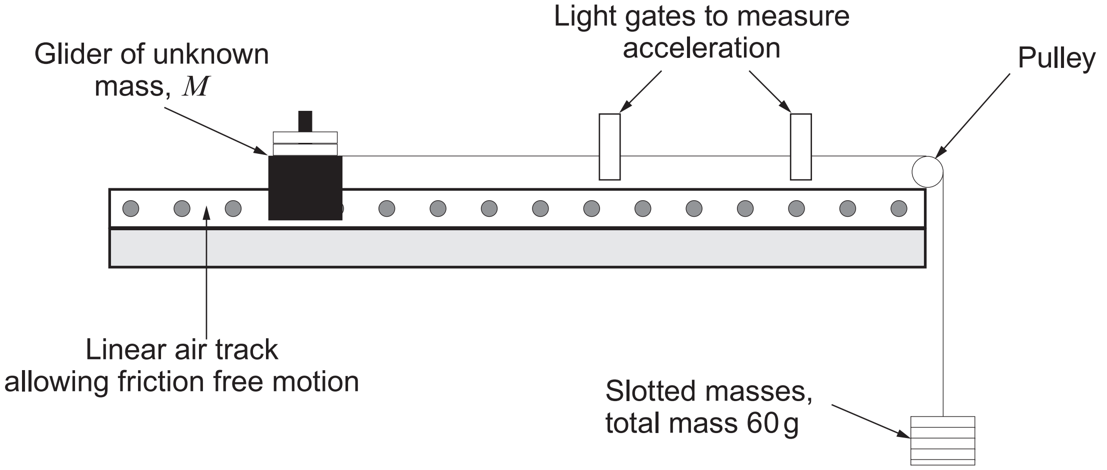
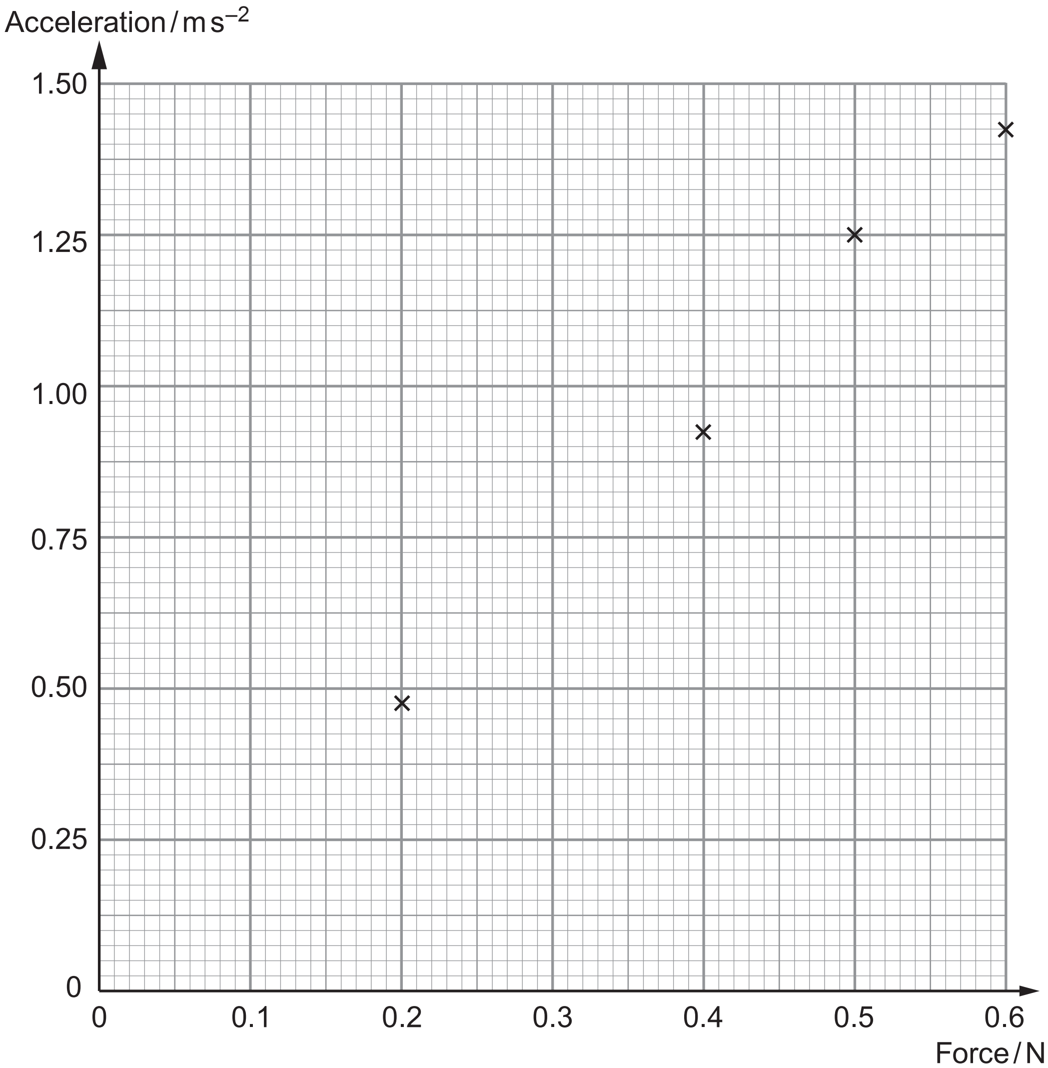
Examiner only

3. *(a)* A student uses the following apparatus to determine the unknown mass, M, of a glider.



The total mass of the system is kept constant by removing one of the slotted masses from the hanging weight and placing it on the glider for each reading of force and acceleration.

She plots **some** of her measurements on the grid below.



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(i) Draw a line of best fit on the graph and use it to determine the gradient. [2]

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(ii) Show clearly that the gradient has units **kg^{-1}**. [2]

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(iii) Determine the value of M, the unknown mass of the glider. [3]

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(iv) Comment on the quality and sufficiency of the data obtained. [2]

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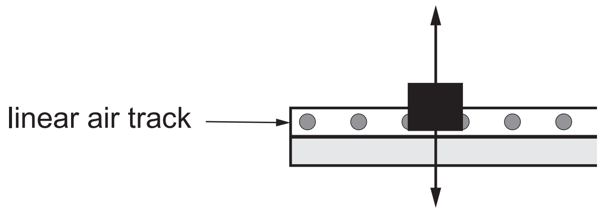
(b) Two vertical forces acting on the glider are shown below.

Force on glider due to air = ... N



Weight of glider = ... N

(i) **Fill in the missing values** on the above diagram. [2]

(ii) The two forces shown **do not** form a Newton 3rd law pair. Give two reasons why. [2]

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